tea for Stata users — a five-minute quick start

You already know how to use this program. This document proves it.

tea (tiny econometric assistant) is a free, open-source (GPLv3), single-binary program that covers the slice of Stata applied economists use daily — import, clean, generate, reshape, regress, xtreg, ivregress, logit, poisson, arima, publication tables, plots — with Stata's syntax and, where it matters, Stata's exact semantics. It runs on Linux and macOS, and **in any web browser with nothing installed**. It ships with six serious practice datasets embedded in the binary, including the complete IMF World Economic Outlook database.

No license server. No installation on locked-down machines. No telemetry — in the browser edition, your data never leaves the tab.

Sixty seconds, zero installation

Open **https://micomrkaic.github.io/tea/** and type:

```
sysuse weo
keep if aggregate==0
xtset iso year
xtreg ngdp_rpch ggxwdg_ngdp if year<=2025, fe
```

That is the complete April-2026 World Economic Outlook — 197 economies, 145 indicators — and a growth-on-debt fixed-effects regression across 193 of them. The output is exactly what your eyes expect:

```
Fixed-effects (within) regression              Number of obs    =     5854
Group variable: iso                           Number of groups =      193

R-sq:                                          Obs per group:
     within  = 0.0132                                 min =       14
     between = 0.0111                                 avg =     30.3
     overall = 0.0115                                 max =       46

--------------------------------------------------------------------------
------
  ngdp_rpch | Coefficient  Std. err.    t    P>|t|    [95% conf. interval]
-------------+------------------------------------------------------------
------
 ggxwdg_ngdp | -0.0201611  0.00231467   -8.71 0.000   -0.0246987  -0.0156234
-------------+------------------------------------------------------------
------
     sigma_u |    1.74613
     sigma_e |    5.44188
         rho |  0.0933459   (fraction of variance due to u_i)
--------------------------------------------------------------------------
------
```

Add `scatter pcpipch lur if year==2024` and a publication-grade vector plot appears beside the terminal. Total elapsed time: about a minute, none of it spent on IT tickets.

Your muscle memory transfers

These work exactly as your fingers already type them:

```
generate lgdp = ln(gdp)
replace  lgdp = . if gdp <= 0
egen     mgdp = mean(gdp), by(country)
bysort country (year): gen g = gdp/gdp[_n-1] - 1
keep if year >= 2000 & !missing(gdp)
sort country year
reshape long gdp, i(country) j(year)
merge 1:1 id using other.dta
tsset year
xtset country year
regress y L.x D2.z i.region c.x#i.region [aw=pop], cluster(country)
ivregress 2sls y (x = z1 z2) w
logit employed education experience
poisson visits chronic income
arima y, arima(1 0 1)
predict yhat
estimates store m1
estimates table m1 m2, se
estout m1 m2 using results.tex
display _b[x] " (" _se[x] ")"
foreach v in gdp inflation unemp {
    quietly summarize `v'
    display "`v': " r(mean)
}
```

The semantics are faithful where faithfulness matters most:

- **Missing values** follow Stata's algebra exactly — including the classic footgun that `if x > 5` *includes* missing, because missing sorts above every number. Your defensive habits (`& !missing(x)`) work unchanged.
- **`by:`** errors on unsorted data, exactly like Stata; `bysort` sorts first; `by g (s):` groups by `g` sorting within by `s`.
- **Time-series operators** `L. F. D. S.` are gap-aware against the declared panel, not row-shifts.
- **Factor variables** `i. c.` # ## expand in estimation varlists with Stata's coefficient naming.
- Weights (`fw aw pw iw`), `robust`, `cluster()`, `if`/`in` — all where you expect them.

Batteries included: sysuse with real data

```
. sysuse dir
   grunfeld   Grunfeld (1958) investment panel: 10 US firms x 1935-
1954 (xtreg, hausman)
  airline  Box-Jenkins airline passengers, monthly 1949-1960 (tsset, arima)
  longley  Longley (1967) US macro, 16 obs: famously ill-conditioned (regress)
  nmes1988  Deb-Trivedi (1997) medical care demand, 4406 persons 66+ (poisson, logit)
   pwt          Penn World Table 10.0 sample: 22 countries x 1950-
2019, CC BY 4.0 (growth)
  weo     IMF World Economic Outlook, April 2026: 197 countries + 13 aggregates, 145 indica
```

The datasets live *inside* the binary — email one file to a colleague and they have the program *and* the data. They are chosen so the classics reproduce:

```
sysuse grunfeld
xtset firm year
quietly xtreg invest value capital, fe
estimates store fe
quietly xtreg invest value capital, re
estimates store re
hausman
```

```
             |      fe            re          Difference     S.E.
-------------+------------------------------------------------------------
------
       value |   0.110124      0.109785     0.000338967    0.00549325
     capital |   0.310065      0.308146      0.00191976    0.00247942
--------------------------------------------------------------------------
------
Test: Ho:  difference in coefficients not systematic
              chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B)
                       =     2.07
            Prob > chi2 =   0.3551
```

Those are the textbook Grunfeld numbers, to the digit. `longley` reproduces the certified solution of the famously ill-conditioned regression; `nmes1988` is real health-economics microdata for count and binary models; the WEO's World and regional aggregates are one `keep if aggregate==1` away.

Three things Stata does not give you

1. **A browser edition.** The identical program compiled to WebAssembly: same commands, same estimators, same data, same output to the byte. Send a colleague a link instead of an installation guide. Uploaded `.dta`/`.csv` files stay in the tab — there is no server.
2. **Reproducibility as a promise.** The same do-file prints *byte-identical* output on every machine, CPU, and BLAS library — engineered (degenerate-case zeros are handled deterministically) and verified by running the entire regres-

sion suite on two maximally different numerical stacks. Referee asks “does this replicate?” — the answer is `diff`.

3. **A single 4 MB binary, free forever.** GPLv3. No license file, no seat count, no expiry, readable source on GitHub.

Honesty section: what tea is not

tea is a deliberately small tool, not a Stata replacement. It has no `mixed`, no `gmm`, no Bayesian suite, no survey (`svy:`) machinery, no `twoway` graphics grammar, no matrix language (`mata`), and `merge m:m` is refused on purpose. The full differences list ships as `COMPATIBILITY.md`. For anything outside scope, the escape hatch is built in:

```
export delimited handoff.csv
shell python3 analyze.py handoff.csv
```

If your day is 80% data-prep and standard estimators — which is most applied days — tea covers it. For the other 20%, it hands off cleanly.

Getting it

- **Try now, install nothing:** https://micomrkaic.github.io/tea/
- **Native binary:**

  ```
  git clone https://github.com/micomrkaic/tea
  cd tea && make && make test     # 42/42 expected
  ```

- **Manual:** `tea-v1.2-manual.pdf` — 38 pages, full command and function reference, generated from the program itself so it cannot drift from the implementation.

Found a bug, or a missing feature you cannot live without? Open an issue — this project moves fast, and field reports drive it.